

Portfolio Optimization I

This document and others to follow present a comprehensive view of a portfolio with four stocks from 01 January 2026 to 31 May 2026. It then forecast the stock prices until 29 June 2026 using monte carlo simulation with 500 simulations and focusing on the 50th percentile simulated path.

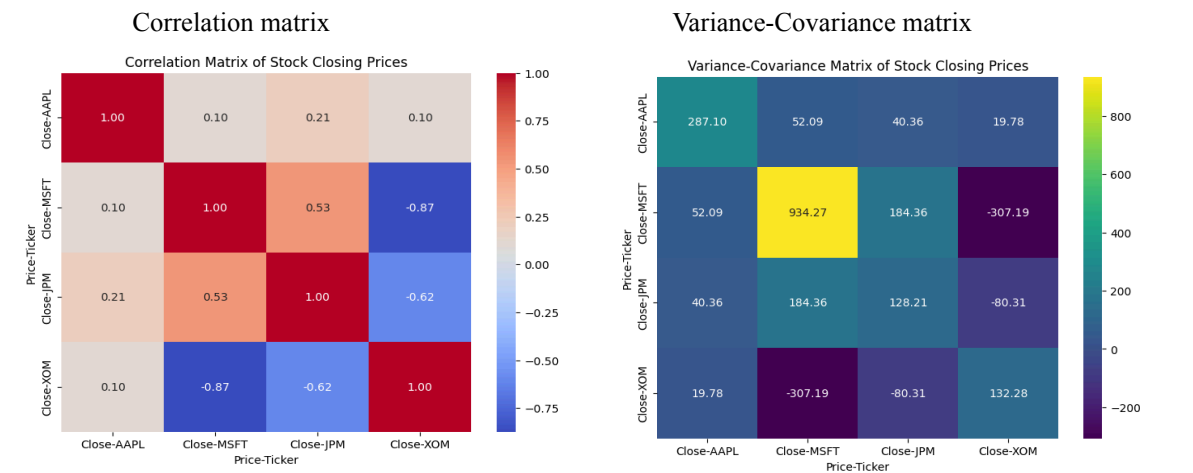
The four stocks used are:

- Apple Inc (AAPL)
- Microsoft Corporation (MSFT)
- JP Morgan Chase & Co. (JPM)
- Exxon Mobil Corporation (XOM)

Sample Statistics

Original data					Forecasted data				
Ticker	AAPL	MSFT	JPM	XOM		AAPL	MSFT	JPM	XOM
count	102.000000	102.000000	102.000000	102.000000	count	133.000000	133.000000	133.000000	133.000000
mean	268.209262	413.711871	303.414902	147.480394	mean	279.707239	423.176591	303.786469	147.451431
std	16.943947	30.565752	11.323081	11.501362	std	25.699911	31.831584	10.064273	10.079029
min	246.242508	355.998932	281.399872	116.911911	min	246.242508	355.998932	281.399872	116.911911
25%	257.350037	396.678352	296.166817	145.308182	25%	258.889008	399.734222	298.780914	145.919101
50%	263.332382	410.411102	303.666260	149.470474	50%	270.460815	419.089996	304.332520	148.396373
75%	273.840218	423.662384	310.205009	153.783699	75%	310.850006	453.258937	309.316986	151.720200
max	312.510010	481.326508	332.906250	170.313995	max	322.960013	481.326508	332.906250	170.313995

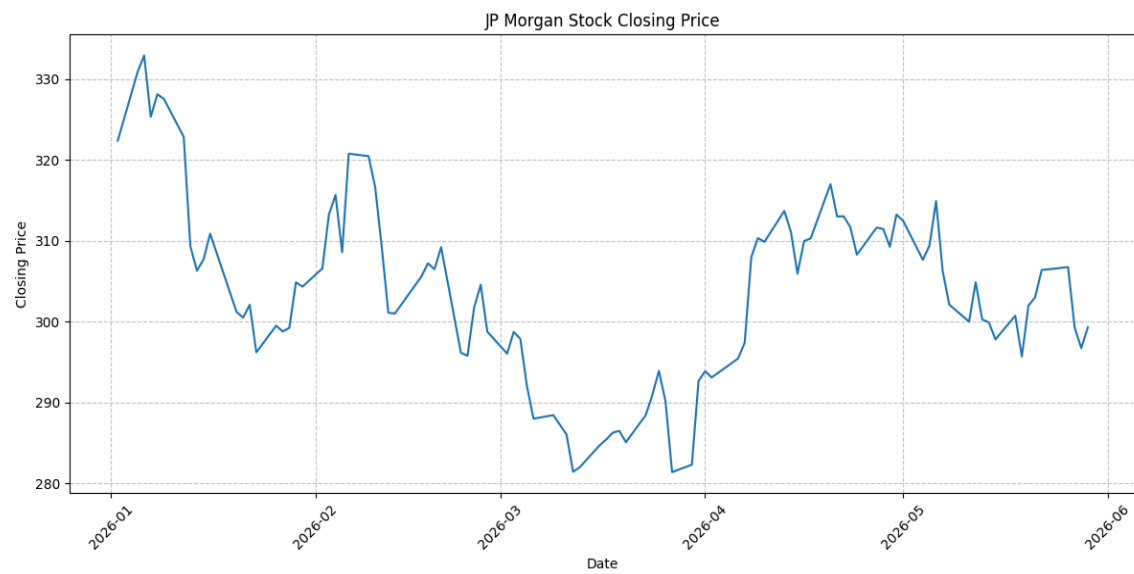
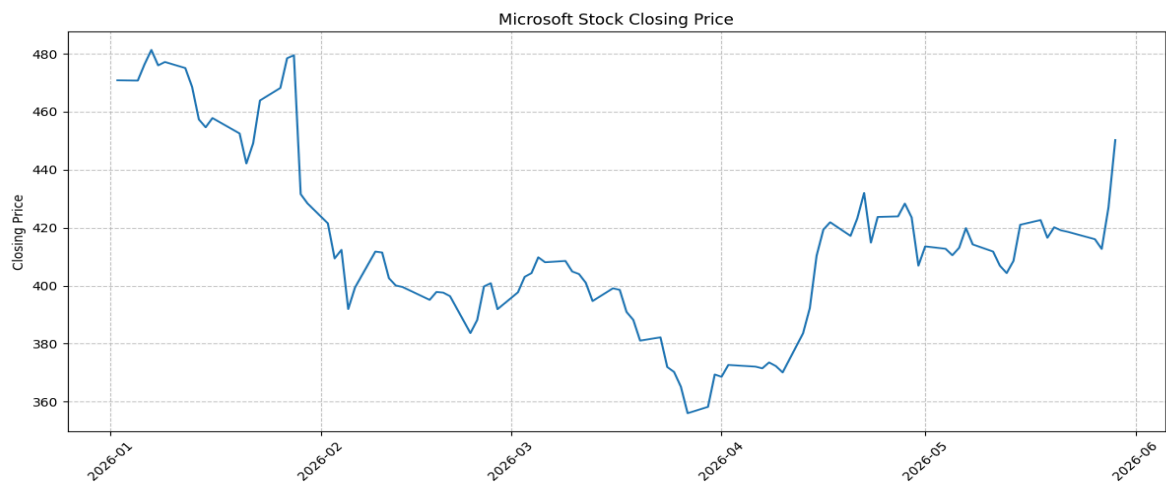
- The mean and variance have remained relatively the same for both data showing that the forecasts do not deviate largely from the original path.

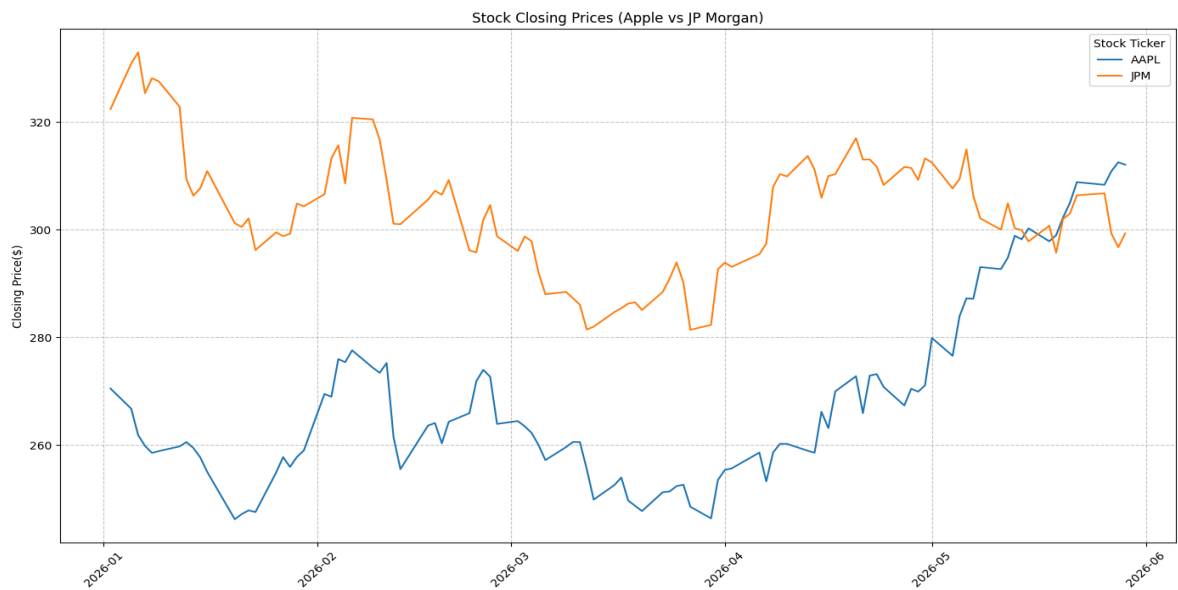
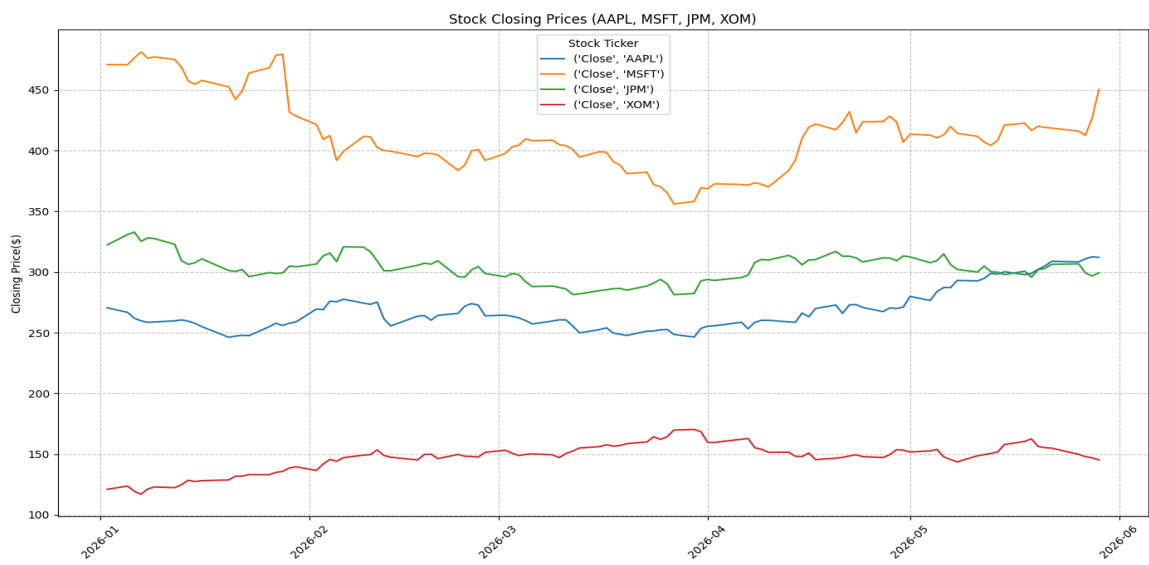
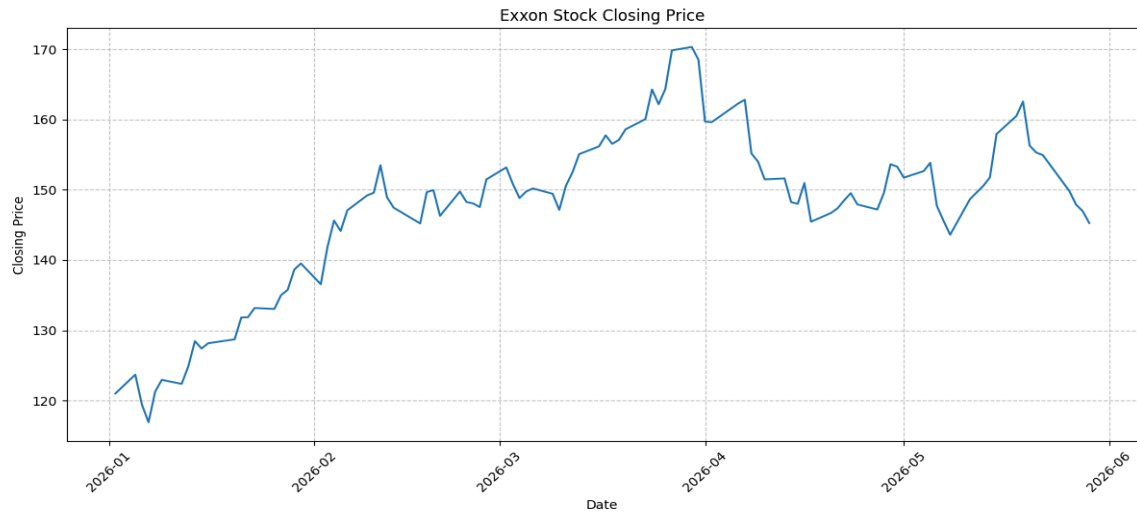


- The main notable correlations are between MSFT & XOM at -0.87 and JPM & XOM at -0.62. Apple is mostly not significantly correlated to any stocks.

- The highest covariances are between MSFT & XOM at -307.19 and MSFT & JPM at 184.36. These high magnitude values are worth attention for comparison to post forecast data.

Stock Charts



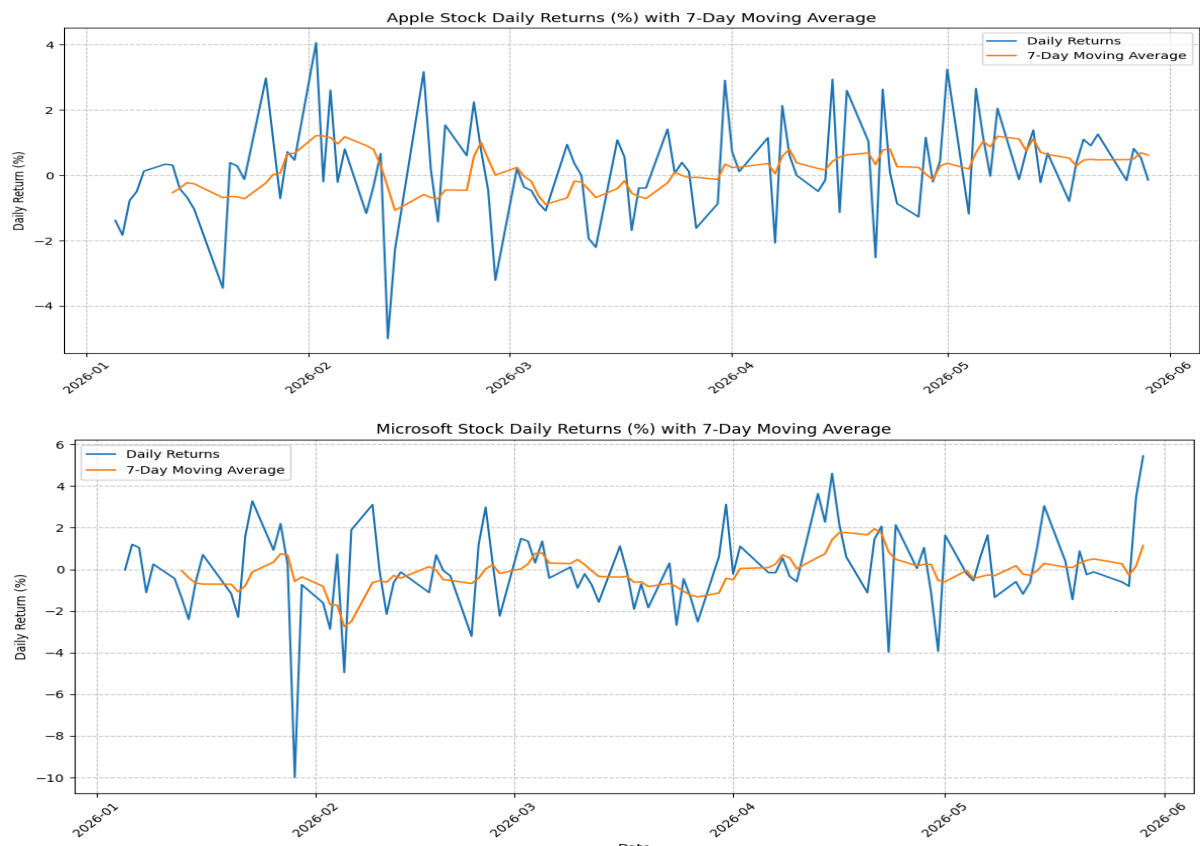


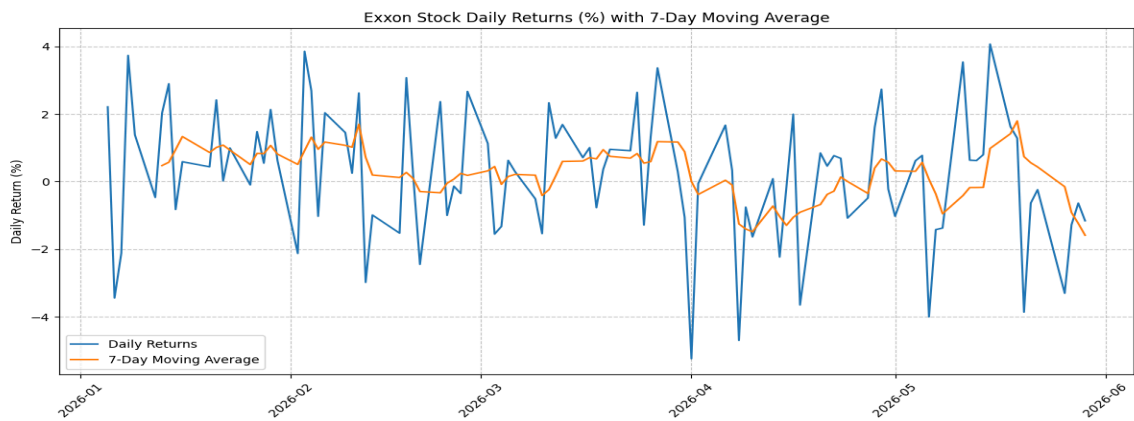
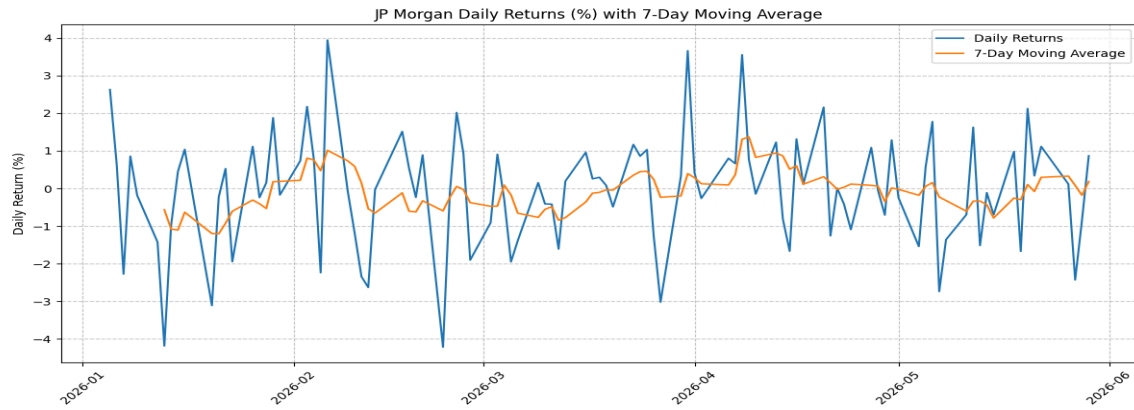


- The stocks are relatively non-stationary and display a lot of turbulence over the 2026 trading year. Apple and JP Morgan stocks are relatively comparable in terms of price range but Apple appears to be showing much steadier growth while long-term (6-months) they range around a similar average, \$312 for Apple and \$332 for JP Morgan.

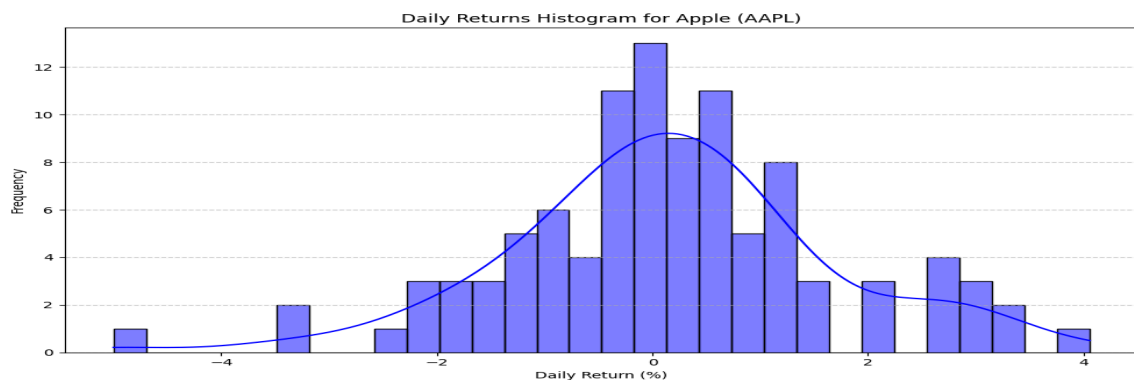
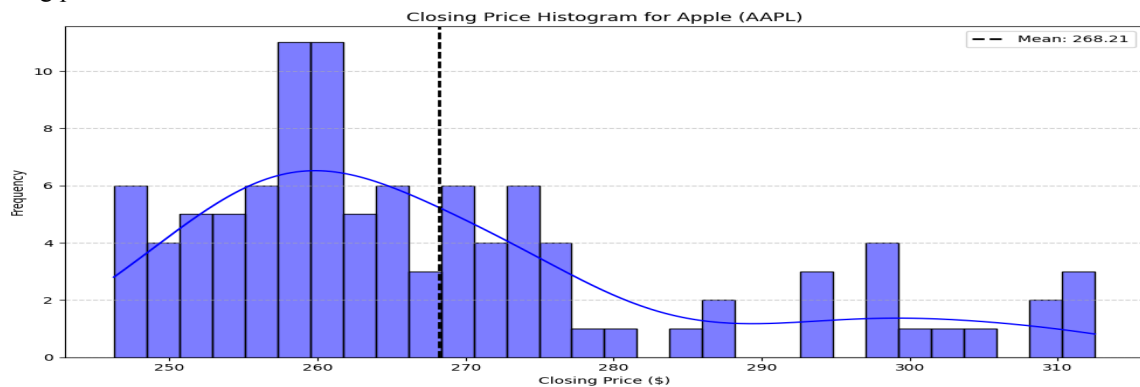
Volatility Plots

Here the returns are calculated as daily percentage change.





- The stocks display significant random noise however for Apple the stock has yielded positive returns since the start of April till the end of May. The rest of the stocks have been highly volatile over the 5-month trading period. The following histogram shows the Apple distributions for closing price and returns over the specified trading period.



Closing Price Distributions

The following shows how the prices are distributed in the 5-months, detailing how each stock price could be shifting over time.

